Vidhisha Balachandran

Senior Researcher, Microsoft Research

vidhishanair.github.io | vidhishanair | vidhishanair@gmail.com | +1-412-961-2637

Summary

My research focuses on efficient post-training methods (SFT, RL) for improving reasoning and agentic capabilities in small language models. I have also built rigorous evaluation frameworks for fine-grained understanding of LLMs. I have been a core contributor to the Phi-4-Reasoning and Eureka ML Insights projects.

Education

2017–2024 **PhD & MSc in Language Technologies**, Carnegie Mellon University (GPA: 4.08/4.0)
PhD Advisor: Prof Yulia Tsvetkov
MSc Advisors: Prof Jaime Carbonell and Prof William Cohen
Thesis: Designing Transparent and Factual Text Generation Systems Grounded in Language Structure

2011–2015 **Bachelors in Computer Science**, PES Institute of Technology (GPA: 9.6/10)
MHRD Scholarship, Distinction Award Recipient

Work Experience

Senior Researcher, Microsoft Research April 2024 - Present

- Conducting multi-turn RL on synthetic coding and terminal-based environments
  - Designing scalable pipelines for generating high-quality synthetic tasks and environments for agent training
  - Developing large-scale RL training recipes for improving SLM performance on TerminalBench, including research on data curriculum, mixtures, and training stability
- Developed RL training and inference-time pruning recipes improving math and science reasoning accuracy by 3–5% while reducing inference token cost by up to 70%.
- Developed Eureka - a framework for rigorous, reproducible model evaluation and insights. (See Eureka ML Insights Repo) and contributed diverse evaluations and insights for Phi-4-Reasoning.

Research Intern, Allen Institute of Artificial Intelligence May 2021 - Aug 2021

(Hosts: Dr Matthew Peters, Dr Pradeep Dasigi)

- Extended Transformer architectures to process documents of ∼16K token length
- Developed pretraining techniques to encourage long range dependencies in representations

Research Intern, Google Brain May 2020 - Aug 2020

(Hosts: Niki Parmar, Dr Ashish Vaswani)

- Developed Scalable Open-Domain QA system with Retrieval Augmented Generation models
- Implemented efficient passage reranking to scale to ∼13M Web documents for fast Top-K MIPS Search

Research Intern, Google Research June 2019 - Aug 2019

(Hosts: Dr William Cohen, Dr Michael Collins)

- Developed KBQA models to reason using Text + Wikidata Background Knowledge
- Trained Fact-Aware text representations for different downstream tasks

Software Development Engineer 2, Flipkart Pvt Limited July 2015 - July 2017

- Developed regression models to produce quality scores for E-Commerce products and listings
- Built a scalable platform for scoring ∼150M entities with low latency

Technical Reports

Abdin M, Agarwal A, **Balachandran V**, Nushi B, ..., Yousefi S, Zheng G. *Phi-4-reasoning Technical Report.* Arxiv 2025

Balachandran V, Nushi B, Chen L, ... Wu Y, Yousefi S. *Inference-Time Scaling for Complex Tasks: Where We Stand and What Lies Ahead.* Arxiv 2025

Balachandran V, Joshi N, Nushi B, ... Yousefi S. *Eureka: Evaluating and Understanding Large Foundation Models.* Arxiv 2024

Select Publications

Vilas MG, ..., Horvitz E, **Balachandran V**. *Tracing the Traces: Latent Temporal Signals for Efficient and Accurate Reasoning.* ICLR 2026

Shrivastava V, Awadallah A, **Balachandran V**, ... Papailiopoulos D. *Sample More to Think Less: Group Filtered Policy Optimization for Concise Reasoning* ICLR 2026

Joshi S, Nushi B, **Balachandran V**, ..., Mirzasoleiman B. *MM-Gen: Principled and Generalizable Data Curation for Enhancing Task Performance in VLMs.* DMLR 2025

Butt N, Chandrasekaran V, Joshi N, Nushi B, **Balachandran V**. *BenchAgents: Multi-Agent Systems for Structured Benchmark Creation* DATA-FM 2024

Brahman F, Kumar S, **Balachandran V**, Dasigi P, ... Hajishirzi H. *The Art of Saying No: Contextual Noncompliance in Language Models.* NeurIPS 2024

Mishra A, Asai A, **Balachandran V**, Wang Y, Neubig G, Tsvetkov Y, Hajishirzi H. *Fine-grained Hallucination Detection and Editing for Language Models.* COLM 2024

Feng S, Shi W, Wang Y, Ding W, **Balachandran V**, Tsvetkov Y. *Don't Hallucinate, Abstain: Identifying LLM Knowledge Gaps via Multi-LLM Collaboration.* ACL 2024

Balachandran V, Hajishirzi H, Cohen W, Tsvetkov Y. *Correcting Diverse Factual Errors in Abstractive Summarization via Post-Editing and Language Model Infilling.* EMNLP 2022

Honours & Awards

Outstanding Paper Award, Area Chair Award-QA Track	ACL 2024
EECS Rising Star	RisingStars 2023
Cadence Technology Scholarship	2022
Google Anita Borg Memorial Award Asia Pacific	2014

Invited Talks

Information Reliability in Large Language Models – Ethics in AI Course, UW	March 2025
Eureka: Evaluating and Understanding Large Foundation Models – NIST	October 2024
Eureka: Evaluating and Understanding Large Foundation Models – MSR Cambridge	September 2024

Technical Skills

Programming Languages	Python (expert), Java (intermediate)
ML/AI	PyTorch, Tensorflow, VeRL, SkyRL, vLLM

References

Dr. Yulia Tsvetkov	Professor, University of Washington	yuliats@cs.washington.edu
Dr. William Cohen	Professor & Principal Scientist, CMU / Google	wcohen@cmu.edu
Dr. Hannaneh Hajishirzi	Professor & Research Manager, UW / AI2	hannaneh@cs.washington.edu